

OCR and Machine Translation Pipeline for Low-Resource Sámi Languages

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Abstract. Northern Sámi is a low-resource language with Optical Character Recognition (OCR) challenges, e.g. limited training data & diacritical characters. In this work, we propose a 2-phase pipeline of OCR & machine translation (MT) comprising lightweight Convolutional Neural Network (CNN) & Connectionist Temporal Classification (CTC) architecture. We compare this to pre-trained Transformer OCR (TrOCR) models from National Library of Norway. The best TrOCR (~350MB) model gets 91.56% character accuracy. Our CNN-CTC model follows this with 87.72% accuracy on 1048 samples, needing only 23MB, and is thus practical for resource-constrained deployment. End-to-end evaluation on 48 parallel sentences reveals viable OCR in our model.

Keywords: OCR · MT · Low-Resource Languages · Sámi · DL · CTC

1 Introduction

Sámi languages are endangered groups of Uralic languages spoken by indigenous people in northern Norway, Sweden, Finland & Russia. Northern Sámi is most widely-spoken (~25,000 speakers). Yet many historical texts are at risk due to lack of digitization & OCR challenges, especially special diacritical characters (Á/á, Č/č, Đ/đ, Ǫ/ǫ, Š/š, Ʒ/Ʒ, e.g. Fig. 1). Most OCR research focuses on high-resource languages, leaving Sámi underrepresented. Commercial systems perform poorly on Sámi orthography [1]. Transformer models e.g. TrOCR [2,1] from National Library of Norway do well (8.55% CER) but need 350MB+, limiting deployment in resource-constrained environment.

okta lea dáhpáhus go bodii dárrolaš njuorjovuopmegierragii gos ledje sámít orrume
It once happened that a Norwegian settler came to the place where the Sámi were living at the Njuorjovuopme uplands

Fig. 1: Example sentence from Johan Turi’s “Muitalus sámiiid birra” [4] with translation.

We propose a novel solution of *CNN-CTC for Sámi*, with comparative study on sentence-length accuracy etc. It is motivated by related work, e.g. [3].

2 Methodology, Data and Architecture

In our proposed *CNN-CTC for Sámi* model, we use the Språkbanken synthetic Sámi OCR data [1] for training. Evaluation occurs with 2 distinct sets: **a)** Språkbanken validation split (1000 synthetic line images) for OCR benchmarking; **b)** 48 parallel handpicked Northern Sámi–English sentence pairs in Johan Turi’s “Muitalus sámiiid birra” [4]. Our CNN-CTC system architecture [5] has 7-layer CNN, bidirectional LSTM encoder (256 units), CTC decoder 32×800px inputs, trained on 276k samples with 395-character vocabulary. VGG-16/19, ResNet-50

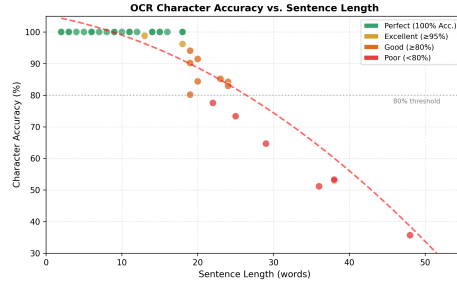


Fig. 2: Character accuracy v. sentence length

backbones fail to converge, and uses CTC loss. The proposed pipeline thus offers a CLI tool to specify images and select an OCR model. It runs translation of TartuNLP neural MT API [6], on completion returning benchmark data in metrics: CER, WER (OCR); BLEU, chrF, TER (MT).

3 Results and Discussion

Table 1 compares TrOCR variants and our CNN-CTC model on the benchmark. The best transformer (trocr_smi_synth) reaches 91.56% character accuracy but requires $\sim 350\text{MB}$ and 12.41s per image. Our ctc_simple trails it by 3.84 accuracy points while being roughly $15\times$ smaller and $380\times$ faster, a favourable trade-off for resource-constrained environment.

Table 1: OCR benchmark on 1048 samples.

Model	Type	Char. Acc. (%)	Word Acc. (%)	s/img	Size
trocr_smi_synth*	Transf.	91.56	72.49	12.41	350MB
ctc_simple (ours)	CNN	87.72	62.45	0.033	23MB
trocr_smi_pred_synth*	Transf.	84.15	54.24	12.08	350MB
trocr_smi	Transf.	75.88	36.50	16.23	350MB

Char./Word Acc. = $100 - \text{CER/WER}$ (normalized Levenshtein).

* smi=Sámi data, pred=+auto-transcribed, synth=+synthetic pre-training.

Sentence Length. Fig. 2 shows that character accuracy is stable (mean $\sim 88\%$, median $>95\%$) for sentences up to ~ 30 words and drops sharply ($\sim 50\%$ mean) for longer sentences ($n = 7$ in this bin). We attribute this to horizontal compression at the fixed $32\times 800\text{px}$ input resolution.

End-to-End Pipeline. On 48 parallel sentence pairs, a full OCR(ctc_simple) $\rightarrow$ MT pipeline yields BLEU 10.40, chrF 33.85, and TER 76.30. While 62.5% of inputs achieve a perfect normalized match (after removing punctuation and collapsing whitespace), overall translation quality remains limited (BLEU ~ 10), indicating that MT quality is the primary constraint on end-to-end performance, though OCR errors ($\sim 88\%$ avg. character accuracy on the book subset) contribute to pipeline limitations.

4 Conclusion and Future Work

Our work in this paper offers a solution for digitizing Sámi corpora, with 2 OCR options: **1. Transformer** (91.56% char. accuracy, 350MB, 12.41s/img): if accuracy is paramount. **2. CNN-CTC** (87.72% char. accuracy, 23MB, 0.033s/img): if footprint and speed matter, giving 3.84 accuracy points for a $\sim 15\times$ smaller and $380\times$ faster system that trains on modest hardware.

Future work: 1. Expand training corpus with real samples to bridge domain gaps. 2. Add automated layout analysis, remove reliance on pre-segmented text.

Acknowledgments and Appendix

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Code: <https://github.com/magwrap/sami-ocr-translate> (access upon request).

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